CoordiWorld: Provenance-Aware Structured World Modeling for Auditable Fixed Candidate-Set Trajectory Evaluation in Autonomous Driving

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***Abstract*—Trajectory evaluation in autonomous driving is inherently counterfactual: the system must assess how a scene may evolve under different ego actions rather than fit a single nominal future. Existing driving world models and plan evaluators are often built in latent or bird's-eye-view (BEV) feature spaces, which are efficient but make decision-time risk auditing, uncertainty calibration, and entity-level explanation difficult. This article presents CoordiWorld, a provenance-aware structured world model for auditable fixed candidate-set trajectory evaluation. CoordiWorld rolls a conflict-resolved structured scene state forward under candidate ego trajectories using explicit ego, agent, and map tokens, and converts the imagined futures into calibrated collision risk, map-grounded rule-violation risk, predictive uncertainty, and post-hoc entity-level attribution. The model is trained in two stages by combining structured rollout learning with pairwise plan-ranking supervision, thereby aligning future prediction with downstream candidate selection. On NAVSIM-v2-aligned shared-candidate trajectory-evaluation benchmarks, CoordiWorld improves the Extended Predictive Driver Model Score (EPDMS) by 1.1 points over the strongest same-protocol baseline, reduces top-1 selected-plan collision and violation rates, and achieves lower calibration error together with stronger auditability metrics, including Risk Drop@1 and Entity Recall@3. These results suggest that structured, entity-grounded future evaluation is a practical alternative to opaque latent-space scoring for trustworthy decision support in safety-critical autonomous driving.**

***Index Terms*—Autonomous driving, candidate trajectory ranking, trustworthy decision support, uncertainty quantification, explainable artificial intelligence.**

I. Introduction

AUTONOMOUS driving decision making is inherently counterfactual. At decision time, an autonomous vehicle must compare several locally feasible maneuvers and select the one that yields the safest and most reliable future once multi-agent interactions, map constraints, and upstream perception uncertainty are taken into account. The problem becomes particularly important in connected and cyber-physical driving systems, where downstream safety decisions must remain auditable under heterogeneous evidence sources. Recent work on digital-twin-enabled traffic guidance for autonomous driving further suggests that future driving decisions are increasingly embedded in IoT-rich system-level coordination rather than isolated onboard planning alone [1], [2]. In the fixed candidate-set setting considered in this article, this requirement appears as an evaluator problem rather than a pure generation problem: the system must rank candidate ego trajectories according to their future consequences, not merely regress a single nominal future.

Recent world-model-based planning and trajectory-evaluation methods have made future imagination increasingly practical [3]–[6]. However, many existing evaluators still operate primarily in latent or bird's-eye-view (BEV) feature spaces [5], [7]–[9]. Although such representations are effective for scalable rollout, they provide limited support for three questions that are central to trustworthy decision-time use: what future risk is being predicted, which scene entity is responsible for that risk, and how certain the evaluator is about its judgment. The limitation becomes more consequential when the evaluator must operate on heterogeneous and imperfect upstream evidence, where confidence, provenance, and partial inconsistency can materially affect downstream safety decisions.

To address this problem, we propose CoordiWorld, a provenance-aware structured world model for auditable trajectory evaluation. CoordiWorld operates on a conflict-resolved structured scene interface in which the current scene is represented by explicit ego, agent, and map tokens together with confidence and provenance cues. Conditioned on a candidate ego trajectory, the model rolls the structured scene state forward and converts the imagined future into calibrated collision risk, map-grounded rule-violation risk, and predictive uncertainty for plan ranking. In addition, CoordiWorld provides post-hoc entity-level attribution so that the final ranking result can be audited against concrete scene participants rather than opaque latent features.

Unlike generic driving world models that emphasize unrestricted future synthesis or latent predictive realism, CoordiWorld is designed as an online evaluator for safety-dominant reranking. It separates evaluator quality from proposal quality by operating on a fixed shared candidate set, treats uncertainty and provenance as first-class decision signals, and preserves a direct link from predicted future risk to explicit entities and map rules. This design is therefore more compatible with trustworthy decision support in connected and cyber-physical driving systems, where downstream debugging, safety analysis, and responsibility tracing matter in addition to raw benchmark score.

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Empirically, CoordiWorld achieves the strongest overall Extended Predictive Driver Model Score (EPDMS) on our NAVSIM-v2-metric-aligned shared-candidate evaluator benchmark, improves the strongest same-protocol baseline by 1.1 EPDMS points, reduces the top-1 selected-plan collision and violation rates, and yields better calibration, robustness, and stronger auditability metrics. These findings suggest that explicit structured evaluation can be a practical alternative to opaque latent-space scoring for safety-critical candidate selection.

The main contributions of this article are as follows:

1) We formulate fixed candidate-set trajectory evaluation as structured, action-conditioned future reasoning on a provenance-aware scene interface, thereby shifting the focus from opaque latent scoring to auditable future consequence evaluation.
2) We develop CoordiWorld, which rolls explicit ego, agent, and map tokens forward under candidate actions and transforms the resulting future states into calibrated collision risk, map-grounded rule-violation risk, predictive uncertainty, and post-hoc entity-level attribution.
3) We establish a shared-candidate training and evaluation protocol that combines structured rollout learning with pairwise plan-ranking supervision, and we validate the proposed method through NAVSIM-v2-aligned evaluator benchmarking and downstream transfer experiments.

The rest of this article is organized as follows. Section II reviews related work. Section III presents the structured scene interface, action-conditioned rollout model, and risk-calibrated evaluator. Section IV reports the experimental results and analysis. Section V concludes this article.

II. Related Work

Related work most relevant to this article falls into three categories: planning-oriented scene interfaces and driving reasoning, driving world models for future imagination, and benchmark-driven trajectory evaluation and reliable plan scoring. Together, these directions have moved autonomous driving from perception-centric pipelines toward more predictive and decision-aware systems. However, fixed candidate-set evaluation imposes a more specific requirement: the evaluator must operate on a structured scene interface that can be rolled forward under candidate actions, remain usable under heterogeneous upstream evidence, and expose auditable risk signals at the entity level. This requirement is not fully met by existing methods.

A. *Planning-Oriented Scene Interfaces and Driving Reasoning*

Planning-oriented end-to-end driving has progressively shifted from monolithic policy learning to more explicit scene abstractions. UniAD [10], VAD [11], SparseDrive [12], and GenAD [13] show that planning quality can benefit from unified task organization, vectorized scene representation, sparse scene abstraction, and generative future modeling. These studies demonstrate that the representation delivered to the planner is a major determinant of downstream driving quality and safety. Nevertheless, in most such systems, scene encoding, candidate generation, and final decision making remain tightly coupled inside one end-to-end stack, leaving open the question of how to construct a standalone evaluator for a fixed shared candidate set.

BEV-based multi-sensor perception has also become the dominant scene carrier for autonomous driving. BEVFusion [14], BEVDepth [15], BEVFormer [16], and TransFusion [17] improve multi-view fusion, depth estimation, temporal aggregation, and camera–LiDAR integration. These methods provide strong geometric and semantic priors for downstream prediction and planning, but their outputs are typically optimized for detection, mapping, or dense feature learning rather than for a conflict-resolved and provenance-aware scene state that can be directly rolled forward under candidate ego actions [18].

Another related thread brings language and higher-level reasoning into driving. DriveLM [19], Talk2BEV [20], DriveGPT4 [21], and DriveVLM [22] enrich scene interpretation, explanation, and high-level decision support. However, their interpretability is predominantly semantic or linguistic and is usually attached after perception. They are not designed as action-conditioned, calibrated plan scorers tied directly to future interaction risk. Taken together, these studies motivate a decision-ready scene interface, but they stop short of defining a standalone structured evaluator for online candidate ranking.

B. *Driving World Models for Future Imagination*

Driving world models provide the closest conceptual foundation for our work because they explicitly model future scene evolution under actions. GAIA-1 [3], DriveDreamer [23], DriveDreamer-2 [24], and DriveWorld [4] demonstrate controllable video generation, customized scenario synthesis, and spatiotemporal pretraining from real driving data. Their main emphasis, however, is usually realism, diversity, or pretraining effectiveness rather than decision-time evaluation of a fixed candidate set.

Recent studies move world modeling closer to planning. LAW [9] uses latent world modeling as a self-supervised signal for end-to-end driving, HERMES [7] unifies scene understanding and future generation, and World4Drive [8] employs a latent world model to generate and evaluate multimodal planning trajectories. These methods narrow the gap between imagination and decision making, but several issues remain for our target setting. The imagined future is usually represented in video, latent, or feature space; uncertainty and confidence are often implicit rather than first-class design objects; and when a candidate is rejected, support is limited for identifying the responsible entity or upstream evidence source.

Hence, the key question is no longer whether a system can imagine the future, but how that imagination should be organized for decision-time reranking. The present work targets this gap by using structured scene tokens and explicit risk decomposition rather than opaque latent scoring alone.

C. Trajectory-Evaluation Benchmarks and Reliable Plan Scoring

Benchmark design has increasingly clarified how candidate plans should be evaluated. NAVSIM [25] and Bench2Drive [26] show that plain open-loop displacement error is an insufficient proxy for driving quality, while Pseudo-Simulation [27] improves offline evaluation by increasing correlation with downstream closed-loop behavior. As a result, the evaluator is no longer a simple trajectory regressor but a module whose scores must align with safety, rule compliance, ego progress, and dynamic consistency.

At the method level, recent planning systems increasingly rely on multiple candidate trajectories together with stronger scorers or teacher signals. WoTE [5] is especially relevant because it performs online trajectory evaluation with a BEV world model. Hydra-MDP [28] and Hydra-MDP++ [29] improve multimodal candidate learning via teacher guidance; GoalFlow [30] and DiffusionDrive [31] strengthen candidate generation; and GTRS [6] and DistillDrive [32] improve candidate scoring and multimodal planning distillation. Collectively, these studies confirm that candidate selection quality is now central to end-to-end driving performance. However, most candidate scorers still operate in latent or BEV feature spaces and are optimized primarily for benchmark gains, offering limited support for calibrated and auditable decision analysis.

Reliability-oriented work provides a complementary perspective. CCTR [33] calibrates predictive uncertainty for motion planning, and CUQDS [34] studies conformal uncertainty quantification under distribution shift. Recent work has also begun to model uncertainty directly in end-to-end autonomous driving pipelines [35]. These methods show that uncertainty estimates cannot be assumed reliable even when average prediction accuracy is competitive, but they remain prediction-centric rather than fixed-candidate evaluator-centric.

Closest to our setting are evaluator-oriented methods such as WoTE [5], scorer-centric planning frameworks such as GTRS [6], and context-sensitive evaluation work such as DriveCritic [36]. Nevertheless, to the best of our knowledge, existing methods do not jointly combine a provenance-aware structured scene interface, candidate-conditioned structured rollout, calibrated risk decomposition, and post-hoc entity-level attribution within one evaluator. This is the gap addressed by CoordiWorld.

III. CoordiWorld Framework

This section describes the problem setup, structured scene interface, candidate-conditioned rollout, risk-decomposed trajectory evaluation, post-hoc entity attribution, and the training/inference pipeline of CoordiWorld.

A. Problem Setup and Structured Scene Interface

We assume access to a conflict-resolved, provenance-aware structured scene interface. In the present implementation, this interface is instantiated with SceneSummary. The formulation below only requires a current world state of the form

$$S_t = \{e_t^{\text{ego}}, A_t, M_t\}, \tag{1}$$

where e_t^{ego} is the ego state, $A_t = \{a_i^t\}_{i=1}^N$ is the set of tracked dynamic agents, and M_t is the static map context. Given a short history $S_{t-h:t} = \{S_{t-h}, \dots, S_t\}$ and a candidate ego trajectory τ, CoordiWorld does not treat the whole future scene as fully stochastic. Instead, the candidate trajectory deterministically specifies the future ego motion and the map remains fixed; the stochastic component is the future evolution of tracked agents:

$$p_\theta\left(\hat{A}_{t+1:t+H} \mid S_{t-h:t}, \tau\right), \tag{2}$$

where H is the rollout horizon. The corresponding predicted structured future at step k is written as

$$\hat{S}_{t+k}(\tau) = \left\{e_{t+k}^{\text{ego}}(\tau), \hat{A}_{t+k}(\tau), M_t\right\}. \tag{3}$$

When traffic-light information is available, the signal phase is attached to the relevant conflict-zone token as a short-horizon contextual attribute, while the geometry of the conflict zone remains part of M_t.

In practice, the evaluator operates on a fixed candidate set rather than synthesizing trajectories from scratch. Let $\mathcal{T}_t = \{\tau^{(m)}\}_{m=1}^M$ denote the candidate ego trajectories supplied by an upstream proposal module. CoordiWorld evaluates this shared candidate pool and selects

$$\tau_t^\star = \arg\min_{\tau \in \mathcal{T}_t} J(\tau), \tag{4}$$

where $J(\tau)$ is the plan-level evaluator score defined later in this section. This separation isolates evaluator quality from proposal quality and matches the shared-candidate protocol used in the experiments.

Each tracked agent is represented as

$$a_i^t = (\text{id}_i, c_i, x_i, y_i, \psi_i, v_{x,i}, v_{y,i}, l_i, w_i, q_i, p_i), \tag{5}$$

where id_i is a within-scene track identity key, c_i is the semantic class, (x_i, y_i, ψ_i) and $(v_{x,i}, v_{y,i})$ denote bird's-eye-view pose and velocity in the ego frame, (l_i, w_i) denote box size, q_i collects confidence or uncertainty cues, and p_i encodes provenance information such as sensor lineage and ambiguity flags.

The current scene is serialized into three token types: one ego token, a set of agent tokens, and a set of map tokens. Continuous kinematic attributes are projected by small multilayer perceptrons, while local identity tags, semantic classes, and provenance tags are embedded through learned lookup tables. Before tokenization, raw tracker IDs are remapped to local slot indices within each scene clip so that temporal consistency is preserved without implying cross-scene semantic meaning. The tokenized history is then encoded by a lightweight temporal Transformer. Unless stated otherwise, we retain at most 32 agents within a 50 m region of interest and at most 24 map tokens to keep the evaluator efficient while preserving the most decision-relevant entities.

B. Candidate-Conditioned Structured Rollout

The candidate ego trajectory is represented as a future polyline

$$\tau = \{(x_{t+1}^{\text{ego}}, y_{t+1}^{\text{ego}}, \psi_{t+1}^{\text{ego}}), \dots, (x_{t+H}^{\text{ego}}, y_{t+H}^{\text{ego}}, \psi_{t+H}^{\text{ego}})\}, \tag{6}$$

which is encoded into action tokens $\{u_k\}_{k=1}^{H}$. Conditioned on the encoded scene history and the current action token, the rollout module updates the hidden state of each tracked agent through action-conditioned interaction reasoning over agent and map tokens. From the resulting hidden state $h_{i,k}$, the model predicts residual motion, existence probability, and heteroscedastic position uncertainty:

$$\begin{aligned}&(\Delta\hat{x}_{i,k}, \Delta\hat{y}_{i,k}, \Delta\hat{\psi}_{i,k},\\ &\Delta\hat{v}_{x,i,k}, \Delta\hat{v}_{y,i,k}, \hat{o}_{i,k}, \hat{\Sigma}^{xy}_{i,k}) = f_\theta(h_{i,k}, u_k),\end{aligned} \tag{7}$$

where $\hat{o}_{i,k} \in [0,1]$ is the predicted existence probability and $\hat{\Sigma}^{xy}_{i,k} = \mathrm{diag}(\hat{\sigma}^2_{x,i,k}, \hat{\sigma}^2_{y,i,k})$ is the predicted diagonal covariance of future position.

The next-step state is updated by residual composition:

$$\begin{aligned}&\hat{x}_{i,k} = \hat{x}_{i,k-1} + \Delta\hat{x}_{i,k}, \qquad \hat{y}_{i,k} = \hat{y}_{i,k-1} + \Delta\hat{y}_{i,k},\\ &\hat{\psi}_{i,k} = \hat{\psi}_{i,k-1} + \Delta\hat{\psi}_{i,k}, \qquad \hat{v}_{x,i,k} = \hat{v}_{x,i,k-1} + \Delta\hat{v}_{x,i,k},\\ &\hat{v}_{y,i,k} = \hat{v}_{y,i,k-1} + \Delta\hat{v}_{y,i,k},\end{aligned} \tag{8}$$

with the base state taken from the last observed frame when $k = 1$. The predicted future box $\hat{b}_{i,k}$ is then reconstructed from $(\hat{x}_{i,k}, \hat{y}_{i,k}, \hat{\psi}_{i,k}, l_i, w_i)$, and the predicted future agent set is written as $\hat{A}_{t+k}(\tau) = \{(\hat{b}_{i,k}, \hat{o}_{i,k}, \hat{\Sigma}^{xy}_{i,k})\}_{i=1}^{N}$. In the current formulation, CoordiWorld does not explicitly spawn unseen agents; instead, it focuses on the future evolution of already tracked entities and predicts whether each remains active. This design is deliberate because the target task is candidate-plan evaluation rather than unrestricted scene synthesis.

C. Risk-Decomposed Trajectory Evaluation

From the predicted future structured states $\hat{S}_{t+1:t+H}(\tau)$, CoordiWorld computes three complementary quantities: collision risk $p_{\mathrm{coll}}(\tau)$, map-grounded rule-violation risk $p_{\mathrm{viol}}(\tau)$, and normalized predictive uncertainty $\bar{u}(\tau)$. These quantities are then combined into a single evaluator score for candidate ranking.

Collision risk: For each rollout step k and tracked agent i, we estimate the instantaneous collision likelihood between the ego box induced by τ and the predicted future box of the agent. Let $b^{\mathrm{ego}}_k(\tau)$ denote the ego box at step k under τ, and let $\hat{b}_{i,k}$ denote the predicted box of agent i. A lightweight collision head takes an uncertainty-aware geometric interaction vector constructed from $(b^{\mathrm{ego}}_k(\tau), \hat{b}_{i,k}, \hat{\Sigma}^{xy}_{i,k})$ and outputs a collision logit

$$c_{i,k} = g_{\mathrm{coll}}\left(b^{\mathrm{ego}}_k(\tau), \hat{b}_{i,k}, \hat{\Sigma}^{xy}_{i,k}\right). \tag{9}$$

The interaction vector includes the relative box-center offset, longitudinal and lateral separations in the ego frame, signed box-clearance margin, relative heading difference, relative closing speed, and the diagonal entries of $\hat{\Sigma}^{xy}_{i,k}$. The logit is converted into an existence-aware collision probability by

$$\tilde{p}^{(i,k)}_{\mathrm{coll}} = \hat{o}_{i,k}\, \sigma(c_{i,k}), \tag{10}$$

which prevents agents predicted to disappear from contributing collision risk at full strength. Plan-level collision risk is then aggregated by a noisy-OR rule:

$$p_{\mathrm{coll}}(\tau) = 1 - \prod_{k=1}^{H}\prod_{i=1}^{N}\left(1 - \tilde{p}^{(i,k)}_{\mathrm{coll}}\right). \tag{11}$$

This aggregation matches the semantics of an "any-collision" event over agents and future steps.

Rule-violation risk: Rule-related risk is computed through a map-grounded soft violation estimator rather than a free black-box classifier. For each rollout step k, we evaluate four soft violation terms

$$v^{\mathrm{lane}}_k,\ v^{\mathrm{da}}_k,\ v^{\mathrm{stop}}_k,\ v^{\mathrm{tl}}_k \in [0,1], \tag{12}$$

corresponding to lane departure, off-drivable-area motion, stop-line violation, and traffic-light-related conflict, respectively. Let m^{lane}_k, m^{da}_k, m^{stop}_k, and m^{tl}_k denote the signed clearances between the future ego footprint and the corresponding map constraints. We convert these margins into soft scores by temperature-scaled sigmoids:

$$\begin{aligned}v^{\mathrm{lane}}_k &= \sigma\left(-\frac{m^{\mathrm{lane}}_k}{T_{\mathrm{lane}}}\right), \quad v^{\mathrm{da}}_k = \sigma\left(-\frac{m^{\mathrm{da}}_k}{T_{\mathrm{da}}}\right),\\ v^{\mathrm{stop}}_k &= \sigma\left(-\frac{m^{\mathrm{stop}}_k}{T_{\mathrm{stop}}}\right), \quad v^{\mathrm{tl}}_k = \sigma\left(-\frac{m^{\mathrm{tl}}_k}{T_{\mathrm{tl}}}\right).\end{aligned} \tag{13}$$

Positive margins indicate safe clearance, whereas negative margins indicate boundary crossing or conflict-zone intrusion. For v^{tl}_k, the corresponding signal state is taken from the structured scene interface so that only prohibited intersection entry is penalized.

To obtain a bounded plan-level violation score, we use a softmax-weighted smooth maximum over all rule types and rollout steps. Let $\mathcal{R} = \{\mathrm{lane}, \mathrm{da}, \mathrm{stop}, \mathrm{tl}\}$ and denote the soft score of rule type $r \in \mathcal{R}$ at step k by v^r_k. We define

$$\alpha_{r,k} = \frac{\exp(\beta_{\mathrm{viol}} v^r_k)}{\sum_{r'\in\mathcal{R}}\sum_{k'=1}^{H}\exp(\beta_{\mathrm{viol}} v^{r'}_{k'})}, \tag{14}$$

and

$$p_{\mathrm{viol}}(\tau) = \sum_{r\in\mathcal{R}}\sum_{k=1}^{H}\alpha_{r,k} v^r_k. \tag{15}$$

Because $\{\alpha_{r,k}\}$ form a simplex, $p_{\mathrm{viol}}(\tau) \in [0,1]$. As β_{viol} increases, this aggregation approaches the dominant violation mode while remaining differentiable and map-traceable.

Predictive uncertainty: Besides collision and rule violation, the evaluator must quantify how certain the model is about its predicted future. We therefore define an existence-weighted predictive uncertainty score

$$u(\tau) = \frac{\sum_{k=1}^{H}\sum_{i=1}^{N}\hat{o}_{i,k}\ \mathrm{tr}(\hat{\Sigma}^{xy}_{i,k})}{\sum_{k=1}^{H}\sum_{i=1}^{N}\hat{o}_{i,k} + \varepsilon}, \tag{16}$$

and normalize it by the 95th percentile of validation-set uncertainty values:

$$\bar{u}(\tau) = \min\left(1, \frac{u(\tau)}{u_{95}}\right). \tag{17}$$

This design avoids over-penalizing crowded scenes with many low-existence or irrelevant entities and makes the uncertainty term more comparable across scenes.

Calibration and ranking: Since the evaluator is intended for decision-time use, the two probability-like quantities are post-hoc calibrated before final ranking:

$$\begin{aligned}\bar{p}_{\mathrm{coll}}(\tau) &= \mathrm{Calib}_{\mathrm{coll}}(p_{\mathrm{coll}}(\tau)),\\ \bar{p}_{\mathrm{viol}}(\tau) &= \mathrm{Calib}_{\mathrm{viol}}(p_{\mathrm{viol}}(\tau)).\end{aligned} \tag{18}$$

Here, $\mathrm{Calib}_{\mathrm{coll}}(\cdot)$ and $\mathrm{Calib}_{\mathrm{viol}}(\cdot)$ are one-dimensional monotonic calibration maps fitted on the validation split and frozen before testing. The final plan-level evaluator score is

$$J(\tau) = \lambda_c \bar{p}_{\mathrm{coll}}(\tau) + \lambda_v \bar{p}_{\mathrm{viol}}(\tau) + \lambda_u \bar{u}(\tau), \tag{19}$$

where λ_c, λ_v, and λ_u are nonnegative weights tuned on the validation split. A lower $J(\tau)$ indicates a safer and more reliable candidate trajectory. In the current formulation, route feasibility and basic comfort are handled primarily by the upstream proposal module, so CoordiWorld focuses on safety-dominant reranking within the shared candidate pool.

D. Post-Hoc Entity Attribution

To make the final ranking auditable, we estimate entity-level responsibility through counterfactual masking. For each nearby entity a_i, we remove the tokens corresponding to that entity from the structured input history $S_{t-h:t}$, rerun the candidate-conditioned rollout and evaluation pipeline for the final selected trajectory, and measure the score drop

$$\Delta_i(\tau) = J(\tau) - J(\tau \mid \mathrm{mask}(a_i)) \,. \tag{20}$$

A larger $\Delta_i(\tau)$ indicates that entity a_i contributes more strongly to the predicted risk of trajectory τ. At inference time, we report the top-K entities with the largest attribution scores as supporting risk evidence. Importantly, attribution is used only for post-hoc auditing rather than for ranking itself: the selected trajectory is determined solely by $J(\tau)$, and attribution is computed afterward to expose the most risk-relevant entities. Because this step requires additional masked forward passes, it is applied only to the final selected plan and only over nearby entities.

E. Training and Inference

CoordiWorld is trained in two stages, followed by post-hoc calibration.

Stage I: structured rollout pretraining. During Stage I, the action-token sequence is instantiated by the logged ego future τ^{gt}, so that the rollout model first learns future agent evolution under the actually executed ego motion before being adapted to shared-candidate evaluation in Stage II.

We first pretrain the rollout model using logged future trajectories and future annotations. The pretraining objective combines future-state prediction, motion supervision, existence prediction, and risk supervision:

$$\mathcal{L}_{\mathrm{rollout}} = \mathcal{L}_{xy} + \alpha_{\mathrm{dyn}}\mathcal{L}_{\mathrm{dyn}} + \alpha_{\mathrm{exist}}\mathcal{L}_{\mathrm{exist}} + \alpha_{\mathrm{risk}}\mathcal{L}_{\mathrm{risk}}. \tag{21}$$

We implement $\mathcal{L}_{xy}$ as Gaussian negative log-likelihood on future position, $\mathcal{L}_{\mathrm{dyn}}$ as Huber loss on velocity and yaw, and $\mathcal{L}_{\mathrm{exist}}$ as binary cross-entropy on existence. The risk term supervises the rollout-induced collision and rule-violation heads under the logged ego future:

$$\begin{aligned}\mathcal{L}_{\mathrm{risk}} = \gamma_{\mathrm{coll}}\,\mathrm{BCE}\big(p_{\mathrm{coll}}(\tau^{\mathrm{gt}}), y_{\mathrm{coll}}\big) \\ + \gamma_{\mathrm{viol}}\,\mathrm{BCE}\big(p_{\mathrm{viol}}(\tau^{\mathrm{gt}}), y_{\mathrm{viol}}\big)\,,\end{aligned} \tag{22}$$

where τ^{gt} denotes the logged ego future, and y_{coll} and y_{viol} are binary labels derived from future collision and map-rule events, respectively. Collision supervision is obtained from future ego-agent box intersections within the logged horizon, while rule-related supervision is derived from map-based future checks on drivable-area occupancy, lane departure, stop-line crossing, and traffic-light-state consistency.

Stage II: pairwise ranking fine-tuning. We then fine-tune the model as a plan evaluator on the shared candidate pool. For each candidate τ, we derive an oracle preference tuple

$$r(\tau) = \big(y^{\mathrm{coll}}(\tau),\, y^{\mathrm{viol}}(\tau),\, -s^{\mathrm{psim}}(\tau),\, -s^{\mathrm{prog}}(\tau)\big), \tag{23}$$

where $y^{\mathrm{coll}}(\tau), y^{\mathrm{viol}}(\tau) \in \{0,1\}$, with value 1 indicating that the candidate incurs a collision or a rule violation within the evaluation horizon and value 0 otherwise. Candidates are ordered lexicographically in ascending order of $r(\tau)$; hence collision-free and rule-compliant candidates are always preferred, pseudo-simulation score resolves ties among equally safe candidates, and progress is used only as a late tie-breaker when safety-relevant outcomes are effectively indistinguishable. We write $\tau_i \succ \tau_j$ if $r(\tau_i)$ is lexicographically smaller than $r(\tau_j)$. For each preferred pair (τ_i, τ_j), we apply the margin ranking loss

$$\mathcal{L}_{\mathrm{rank}} = \max(0,\, m - (J(\tau_j) - J(\tau_i)))\,. \tag{24}$$

The full training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{rollout}} + \beta_{\mathrm{rank}}\mathcal{L}_{\mathrm{rank}}. \tag{25}$$

Post-training calibration and inference. After Stage II, the one-dimensional calibration maps in (18) are fitted on the validation split and then frozen for final testing. At inference time, scene encoding is shared across all candidates in the same scene and only the action tokens vary, which allows parallel evaluation of the candidate pool and keeps the evaluator practical for downstream control use.

IV. Experiments

We first describe the experimental setup, including the dataset split, evaluation protocol, baselines, metrics, and implementation details. We then evaluate CoordiWorld from six complementary perspectives: headline benchmark performance, scenario-wise behavior, evaluator quality and reliability, ablation and efficiency, qualitative interpretability, and downstream closed-loop transfer.

A. Experimental Setup

Datasets and protocol: We use the NAVSIM/OpenScene `navtrain` split as the primary training corpus. Unless otherwise stated, all headline quantitative results are reported on a NAVSIM-v2-metric-aligned shared-candidate evaluator benchmark: the official NAVSIM v2 metric suite is applied locally on `navhard two-stage` scenes under our shared-candidate evaluation protocol. The resulting numbers should therefore be interpreted as same-protocol evaluator comparisons rather than as direct leaderboard submissions. Whenever official benchmark results from prior work are discussed, we preserve their original evaluation setting and report them separately. nuScenes is used only for optional auxiliary pretraining and cross-domain analysis [37]. Following NAVSIM

v2, the primary benchmark metric is the Extended Predictive Driver Model Score (EPDMS), which aggregates safety, rule compliance, progress, and comfort under a pseudo closed-loop protocol [25], [27].

To isolate evaluator quality from candidate-generation quality, all same-protocol comparisons follow a shared-candidate design. For each scene, every method evaluates the same causal candidate pool consisting of one nominal proposal, three speed-scaled variants, four lateral-shift variants, and four curvature-perturbed variants. Expert-anchored candidate pools, if used at all, are restricted to controlled validation analyses and are not used in the headline table. In addition, a small closed-loop transfer study is conducted on Bench2Drive whenever the downstream wrapper is available. This transfer benchmark reports Driving Score (DS) and Success Rate (SR) and is used only to verify that better trajectory ordering can transfer to executable control behavior [26]. In all quantitative tables, values report the mean over three random seeds; paired bootstrap resampling over scenes is used to estimate 95% confidence intervals during significance checking.

Baselines: We consider two complementary baseline groups. The main quantitative tables in this article focus on same-protocol shared-candidate comparisons, whereas official NAVSIM benchmark anchors are used only as broader contextual references and, when included, are reported separately from the shared-candidate tables. The first group consists of official benchmark anchors from the NAVSIM ecosystem, including ConstantVelocity, EgoStatusMLP, TransFuser, and LatentTransFuser. These methods are reported only to position CoordiWorld within the broader benchmark landscape rather than as the primary same-interface comparison set. The second group forms the main same-protocol comparison set: (i) a TTC + map-rule heuristic baseline; (ii) a current-state scorer using the same structured-scene input, instantiated with SceneSummary in the present implementation, but without future rollout; (iii) a latent/BEV evaluator-style baseline that performs candidate-conditioned evaluation in an implicit latent or BEV space without explicit entity grounding, provenance modeling, or post-hoc entity attribution; and (iv) a WoTE-style baseline adapted to the same shared-candidate protocol [5]. Methods with the suffix "-style" denote protocol-aligned re-instantiations rather than exact line-by-line reproductions of the original papers. A HERMES-style comparison, when included, is reported in the appendix only because it serves mainly as conceptual context for unified scene understanding and generation rather than as the most direct baseline for fixed-candidate trajectory evaluation [7].

Metrics: The headline benchmark reports EPDMS together with its official NAVSIM-v2 submetrics: No at-fault Collisions (NC), Drivable Area Compliance (DAC), Driving Direction Compliance (DDC), Traffic Light Compliance (TLC), Ego Progress (EP), Time-to-Collision within bound (TTC), Lane Keeping (LK), History Comfort (HC), and Extended Comfort (EC) [25], [27]. To directly evaluate evaluator quality, we additionally report Spearman's ρ, Kendall's τ, and NDCG@3 between evaluator scores and pseudo-simulation outcomes, as well as the top-1 selected-plan collision rate and top-1 selected-plan violation rate. Because CoordiWorld explicitly models confidence and uncertainty, reliability is assessed using Expected Calibration Error (ECE) and Brier Score for both collision risk and rule-violation risk. Robustness under imperfect structured-scene input is measured by ranking stability together with EPDMS drops under confidence noise, provenance masking, and evidence dropout. Here, ranking stability denotes the consistency between the clean and perturbed within-scene candidate rankings; larger values indicate that the evaluator preserves candidate ordering more faithfully under input perturbation.

To evaluate auditability, we further report Risk Drop@K and Entity Recall@K under counterfactual entity masking. Let $\hat{\mathcal{A}}_K(S,\tau)$ denote the top-K entities ranked by attribution score for candidate τ in scene S. For brevity, let $\hat{\mathcal{A}}_K \equiv \hat{\mathcal{A}}_K(S,\tau)$ and $\mathcal{A}^{\mathrm{gt}} \equiv \mathcal{A}^{\mathrm{gt}}(S,\tau)$ whenever the scene–candidate pair (S,τ) is clear from context. We define

$$\mathrm{RiskDrop@}K = \frac{1}{|\mathcal{D}|} \sum_{(S,\tau)\in\mathcal{D}} \frac{J(\tau) - J\Big(\tau \mid \mathrm{mask}\Big(\hat{\mathcal{A}}_K\Big)\Big)}{J(\tau)+\varepsilon}. \tag{26}$$

where higher values indicate that the reported entities explain a larger fraction of the evaluator score. For interactive-risk scenes, let $\mathcal{A}^{\mathrm{gt}}$ denote the ground-truth risky-entity set derived from future collision-partner annotations. We then define

$$\mathrm{EntityRecall@}K = \frac{1}{|\mathcal{D}_{\mathrm{int}}|} \sum_{(S,\tau)\in\mathcal{D}_{\mathrm{int}}} \mathbf{1}\Big[\hat{\mathcal{A}}_K \cap \mathcal{A}^{\mathrm{gt}} \neq \emptyset\Big]. \tag{27}$$

Higher values indicate that the attribution analysis more reliably recovers the truly risk-relevant scene participants. Finally, when the downstream wrapper is available, the closed-loop transfer experiment on Bench2Drive reports Driving Score and Success Rate following the official benchmark protocol [26].

Implementation details: All models are implemented in PyTorch and trained with AdamW using mixed-precision training. The structured scene encoder is a lightweight Transformer operating on structured-scene tokens, which are constructed from SceneSummary in the present implementation. Unless otherwise specified, each scene retains at most 32 dynamic agents and 24 map tokens after spatial filtering. The evaluator encodes 2 s of structured history and scores 4 s candidate trajectories, matching the standard planning horizon used by NAVSIM [25]. Training follows the two-stage procedure described in Section III: rollout pretraining is followed by pairwise ranking fine-tuning, and calibration is fitted post hoc on the validation split and frozen before final testing. To improve robustness, we apply three lightweight perturbations during training: evidence dropout, confidence noise, and provenance masking. Concretely, evidence dropout randomly removes a subset of non-ego evidence tokens, confidence noise perturbs confidence-related fields in the structured-scene input, and provenance masking replaces provenance tags with a null token. In the present implementation, evidence dropout is instantiated as random agent dropout, and confidence noise is instantiated as confidence jitter. All same-protocol learned baselines are trained under the same candidate pool, trajectory horizon, and data split, and their parameter counts are kept within the same order of magnitude whenever possible. Unless

TABLE I
MAIN QUANTITATIVE COMPARISON ON THE NAVSIM-V2-ALIGNED SHARED-CANDIDATE EVALUATOR BENCHMARK. THE OFFICIAL NAVSIM V2 METRIC SUITE IS APPLIED LOCALLY ON NAVHARD TWO-STAGE SCENES UNDER THE SHARED-CANDIDATE PROTOCOL, SO THE TABLE SHOULD BE INTERPRETED AS A SAME-PROTOCOL EVALUATOR COMPARISON RATHER THAN A DIRECT LEADERBOARD SUBMISSION. HIGHER IS BETTER FOR ALL METRICS. VALUES REPORT THE MEAN OVER THREE RANDOM SEEDS.

Method	NC↑	DAC↑	DDC↑	TLC↑	EP↑	TTC↑	LK↑	HC↑	EC↑	EPDMS↑
TTC + map-rule	96.7	93.4	98.3	99.1	82.7	94.6	92.2	97.5	69.4	79.8
Current-state scorer	97.0	94.1	98.6	99.3	83.1	95.0	93.1	97.6	71.2	81.7
Latent/BEV evaluator-style	97.3	94.8	98.8	99.4	83.7	95.6	94.0	97.8	73.6	83.9
WoTE-style	97.4	95.0	98.8	99.5	84.0	95.8	94.3	97.8	74.8	84.8
CoordiWorld (ours)	**97.8**	**95.6**	**99.1**	**99.6**	**84.4**	**96.3**	**95.2**	**98.2**	**77.1**	**85.9**

otherwise stated, latency is reported per scene over the full 12-candidate shared pool, and throughput is measured under the same hardware setting for all methods.

B. Main Results on the NAVSIM-v2-Aligned Shared-Candidate Benchmark

Table I reports the headline comparison on the NAVSIM-v2-aligned shared-candidate evaluator benchmark.

CoordiWorld achieves the highest overall EPDMS of 85.9, improving the strongest same-protocol baseline (WoTE-style) by 1.1 points. Importantly, the gain is not driven by ego progress alone. The improvement is concentrated on NC, TTC, LK, and EC, indicating that the proposed structured evaluator improves candidate selection primarily by suppressing unsafe, dynamically inconsistent, and geometrically implausible trajectories.

Compared with the current-state scorer, CoordiWorld improves EPDMS from 81.7 to 85.9 while also improving every submetric related to safety and stability listed in Table I. This result verifies that the benefit does not come merely from using a structured present-state representation. Rather, the gain arises from candidate-conditioned future evaluation. Compared with the latent/BEV evaluator-style baseline, CoordiWorld also yields a more balanced performance profile, supporting the argument that explicit entity grounding, provenance modeling, and structured rollout supervision are more suitable for decision-time trajectory evaluation than purely implicit latent dynamics.

C. Scenario-Wise Analysis

Representative scenario categories are summarized in Table II, including lane following, lane change, merge/cut-in, unprotected turn, pedestrian crossing, and occlusion-heavy interaction. Scenes are grouped using map priors, interaction tags, and trajectory-template heuristics. The gains of CoordiWorld are smallest on simple lane-following scenes and become progressively larger as interaction complexity increases.

The gains of CoordiWorld are smallest on simple lane-following scenes and become progressively larger as interaction complexity increases. Relative to the strongest baseline, the improvement is only $+0.2$ on lane following, but rises to $+0.9$ on lane change, $+1.7$ on merge/cut-in, $+1.3$ on unprotected turn, $+2.2$ on pedestrian crossing, and $+2.7$ on occlusion-heavy cases. This trend is consistent with the design motivation of the method: structured future evaluation is most valuable when future safety depends on entity-level interaction reasoning rather than on trivial ego-state extrapolation.

TABLE II
SCENARIO-WISE EPDMS ON REPRESENTATIVE SCENARIO CATEGORIES. GAINS ARE MODEST IN SIMPLE LANE-FOLLOWING CASES BUT BECOME LARGER UNDER STRONG INTERACTION AND AMBIGUITY. HIGHER IS BETTER. VALUES REPORT THE MEAN OVER THREE RANDOM SEEDS.

Method	Lane follow	Lane change	Merge/ cut-in	Unprot. turn	Ped. cross.	Occl. heavy
Current-state scorer	86.9	82.8	78.9	79.3	76.8	74.2
Latent/BEV eval.-style	87.3	84.1	80.4	80.5	78.1	75.6
WoTE-style	87.5	84.5	81.0	80.8	78.6	76.3
CoordiWorld (ours)	**87.7**	**85.4**	**82.7**	**82.1**	**80.8**	**79.0**

Figure 1 complements Table II in two ways. The left panel directly visualizes the scenario-wise EPDMS gains computed from Table II. The right panel provides a difficulty-stratified view consistent with the same qualitative pattern: the advantage of CoordiWorld remains limited when the candidate pool is easy, but becomes larger when the evaluator must discriminate among several plausible yet unequally safe actions. This view is consistent with the interpretation that the proposed structured evaluator is most beneficial in interaction-heavy and ambiguity-prone decision regimes.

D. Evaluator Quality, Calibration, and Robustness

Table III focuses on evaluator quality. CoordiWorld achieves the highest Spearman's ρ, Kendall's τ, and NDCG@3, and it also yields the lowest top-1 selected-plan collision and violation rates. This result indicates that better trajectory ordering indeed translates into safer final choices.

Relative to WoTE-style, CoordiWorld improves Spearman's ρ from 0.61 to 0.66, Kendall's τ from 0.46 to 0.50, and NDCG@3 from 0.83 to 0.87, while reducing the top-1 collision rate from 1.6% to 1.3% and the top-1 violation rate from 5.9% to 4.9%. These improvements show that the proposed structured evaluator is not merely strong on the headline benchmark; it is also better aligned with pseudo-simulation outcomes at the candidate-ranking level.

Calibration results are reported in Table IV. CoordiWorld reduces both ECE and Brier Score for collision risk and rule-violation risk, indicating that its risk estimates are not only discriminative but also better calibrated for decision-time use.

Relative to WoTE-style, CoordiWorld reduces collision-risk ECE from 0.083 to 0.058 and rule-violation ECE from 0.078

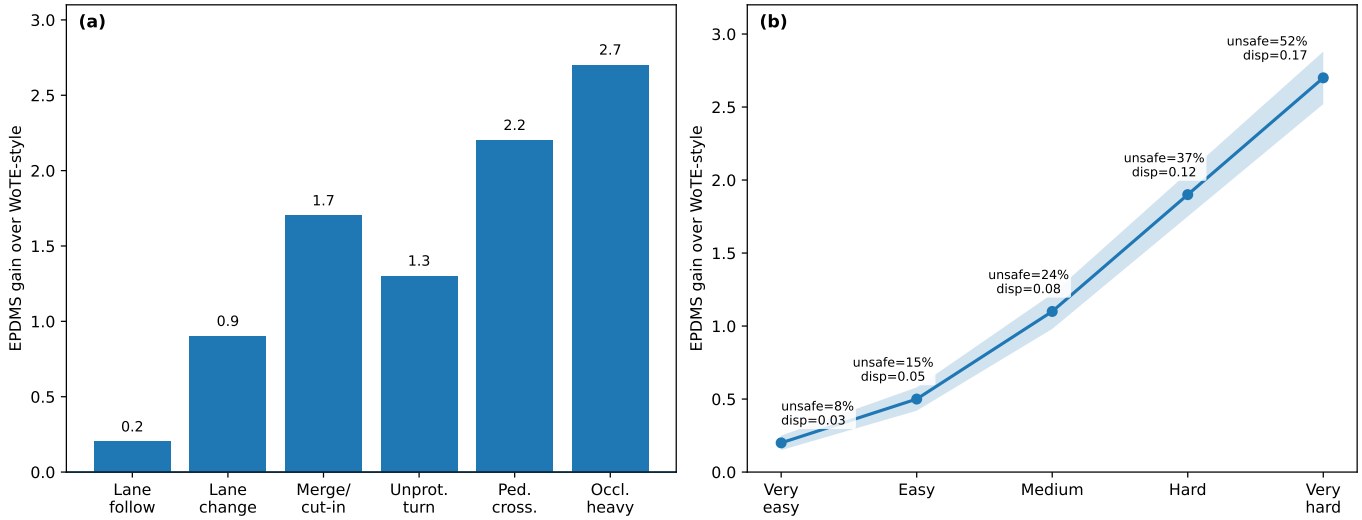

Fig. 1. Scenario-wise and difficulty-stratified analysis. Left: EPDMS gains of CoordiWorld over the strongest baseline across representative scenario types. Right: relative gain as a function of candidate-pool hardness, measured by the proportion of unsafe candidates and the dispersion of trajectory scores. The advantage of CoordiWorld widens as the decision set becomes harder.

TABLE III
EVALUATOR-SPECIFIC METRICS. HIGHER IS BETTER FOR SPEARMAN'S ρ, KENDALL'S τ, AND NDCG@3; LOWER IS BETTER FOR TOP-1 COLLISION AND TOP-1 VIOLATION. VALUES REPORT THE MEAN OVER THREE RANDOM SEEDS.

Method	Spearman's $\rho \uparrow$	Kendall's $\tau \uparrow$	NDCG@3 $\uparrow$	Top-1 coll. $\downarrow$	Top-1 viol. $\downarrow$
TTC + map-rule	0.44	0.32	0.71	2.6%	8.1%
Current-state scorer	0.51	0.38	0.76	2.1%	7.0%
Latent/BEV eval.-style	0.58	0.43	0.81	1.8%	6.3%
WoTE-style	0.61	0.46	0.83	1.6%	5.9%
CoordiWorld (ours)	**0.66**	**0.50**	**0.87**	**1.3%**	**4.9%**

TABLE IV
CALIBRATION METRICS FOR COLLISION RISK AND RULE-VIOLATION RISK. LOWER IS BETTER. VALUES REPORT THE MEAN OVER THREE RANDOM SEEDS.

Method	ECE coll$\downarrow$	Brier coll$\downarrow$	ECE viol$\downarrow$	Brier viol$\downarrow$
Current-state scorer	0.102	0.137	0.094	0.112
Latent/BEV eval.-style	0.091	0.124	0.084	0.101
WoTE-style	0.083	0.116	0.078	0.094
CoordiWorld (ours)	**0.058**	**0.089**	**0.057**	**0.081**

to 0.057. Similar improvements are observed on Brier Score. These reductions are important because ranking candidate trajectories in realistic driving scenes often depends on the reliable estimation of non-extreme but safety-relevant risks.

Figure 2 provides an intuitive visualization consistent with the aggregate calibration results in Table IV. Anchored by the observed ECE and Brier-score differences, the curves illustrate that CoordiWorld remains closer to ideal calibration for both collision risk and rule-violation risk, especially in the medium-risk regime where reliable risk estimation is most important for candidate reranking.

TABLE V
ROBUSTNESS UNDER STRUCTURED-INPUT PERTURBATIONS. HIGHER IS BETTER FOR RANKING STABILITY; LOWER IS BETTER FOR ABSOLUTE EPDMS DROP UNDER PERTURBATION. VALUES REPORT THE MEAN OVER THREE RANDOM SEEDS.

Method	Rank stab. $\uparrow$	ΔEPDMS conf. noise$\downarrow$	ΔEPDMS prov. mask$\downarrow$	ΔEPDMS evid. drop.$\downarrow$
Current-state scorer	0.73	3.4	3.9	3.5
Latent/BEV eval.-style	0.79	2.9	3.4	3.1
WoTE-style	0.81	2.5	3.2	2.8
CoordiWorld (ours)	**0.87**	**1.6**	**1.8**	**1.7**

Robustness under structured-input perturbations is summarized in Table V. To make the direction of improvement unambiguous, we report absolute EPDMS drops under perturbation, so lower values are better. CoordiWorld achieves the highest ranking stability and the smallest degradation under confidence noise, provenance masking, and evidence dropout.

Compared with WoTE-style, CoordiWorld improves ranking stability from 0.81 to 0.87 while reducing the absolute EPDMS drop from 2.5 to 1.6 under confidence noise, from 3.2 to 1.8 under provenance masking, and from 2.8 to 1.7 under evidence dropout. This behavior is especially important because the proposed evaluator is designed to operate on structured summaries produced by imperfect upstream modules, where robustness to partial inconsistency is a practical requirement rather than an optional property.

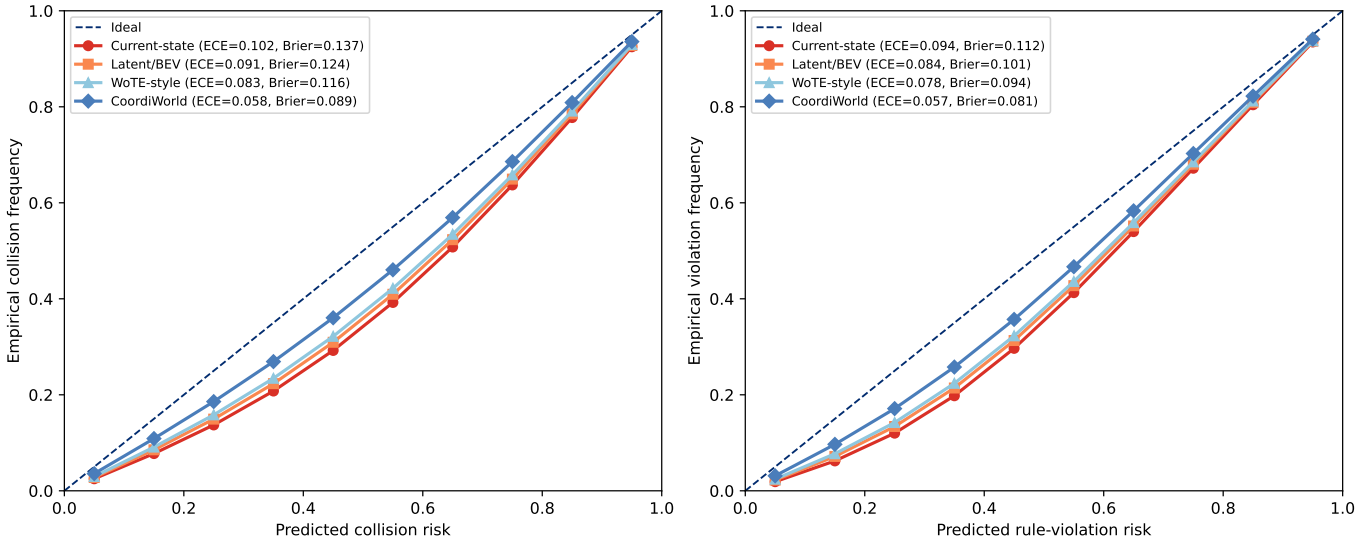

Fig. 2. Reliability visualization for collision risk and rule-violation risk on NAVSIM v2. The plotted curves are anchored by the aggregate calibration results in Table IV and highlight the tighter calibration of CoordiWorld, especially in the medium-risk regime.

TABLE VI
ABLATION STUDY OF THE PROPOSED COMPONENTS. HIGHER IS BETTER FOR EPDMS, SPEARMAN'S ρ, RISK DROP@1, AND ENTITY RECALL@3; LOWER IS BETTER FOR ECE AND LATENCY. FT = FINE-TUNING.

Variant	EPDMS↑	Spearman's ρ ↑	ECE coll↓	Risk Drop@1↑	Entity Recall@3↑	Latency (ms)↓
CoordiWorld (full)	**85.9**	**0.66**	**0.058**	**0.119**	**0.74**	24
w/o struct. tokens	84.5	0.60	0.079	0.072	0.41	22
w/o future rollout	83.8	0.57	0.081	0.069	0.38	**18**
w/o prov./conf.	85.0	0.63	0.074	0.101	0.68	24
w/o calibration	85.2	0.65	0.093	0.116	**0.74**	24
w/o attribution	**85.9**	**0.66**	**0.058**	0.083	0.52	23
w/o pairwise ranking FT	84.8	0.62	0.061	0.108	0.70	24

E. Ablation and Efficiency

Table VI reports the main ablation study. Six components are examined: structured-scene tokens, candidate-conditioned future rollout, provenance/confidence encoding, explicit calibration, post-hoc attribution, and pairwise ranking fine-tuning.

Three observations are especially important. First, removing future rollout causes the largest drop in EPDMS (from 85.9 to 83.8) and rank correlation (from 0.66 to 0.57), showing that the main gain cannot be explained by structured current-state encoding alone. Second, removing explicit structural tokens also leads to a large degradation (EPDMS 84.5; $\rho = 0.60$), confirming that explicit entity structure is crucial to the evaluator. Third, removing provenance/confidence cues weakens both EPDMS and calibration, whereas removing explicit calibration leaves rank correlation nearly unchanged but noticeably worsens ECE. Hence, calibration primarily improves decision-time reliability, while provenance and confidence cues improve both robustness and ranking quality.

Removing the attribution module leaves the main benchmark score unchanged but substantially reduces Risk Drop@1 and Entity Recall@3, which shows that attribution primarily contributes to auditability rather than raw benchmark performance. Removing pairwise ranking fine-tuning also causes a broader quality drop, confirming that second-stage ranking supervision is a meaningful complement to rollout pretraining.

TABLE VII
EFFICIENCY COMPARISON. THROUGHPUT IS MEASURED AS THE NUMBER OF EVALUATED SCENES PER SECOND ON THE SAME HARDWARE CONFIGURATION.

Method	Params (M)↓	GPU mem. (GB)↓	Latency (ms)↓	Throughput ↑
TTC + map-rule	–	0.2	3	333.3
Current-state scorer	8.4	3.1	9	111.1
Latent/BEV eval.-style	14.8	6.5	18	55.6
WoTE-style	16.2	7.4	23	43.5
CoordiWorld (ours)	**17.6**	**7.8**	**24**	**41.7**

A performance–latency view is shown in Fig. 3, while Table VII reports the efficiency statistics explicitly. Figure 3 makes the trade-off visually explicit: CoordiWorld and WoTE-style occupy nearly the same latency regime, yet CoordiWorld shifts the operating point upward in EPDMS and therefore lies on a more favorable Pareto frontier.

Compared with WoTE-style, CoordiWorld introduces only a marginal latency increase (24 ms versus 23 ms) while delivering the strongest headline performance. Table VI further shows that the full design preserves strong auditability metrics, with higher Risk Drop@1 and Entity Recall@3 than the

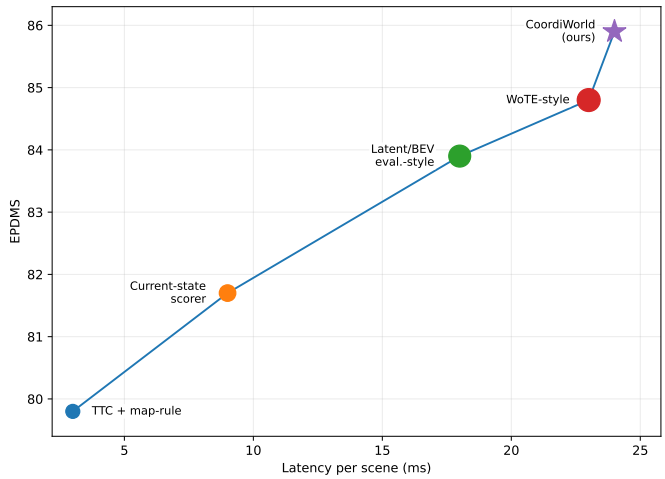

Fig. 3. Performance–latency trade-off of all same-protocol baselines. CoordiWorld lies on a favorable Pareto frontier, delivering the strongest EPDMS with only marginal overhead relative to the strongest evaluator baseline. Marker size is proportional to parameter count.

attribution-removed variant at nearly identical latency. This result supports the positioning of CoordiWorld as a practical decision module rather than as an overly expensive research prototype.

F. Qualitative Analysis

Figure 4 presents three illustrative structured-scene visual summaries of representative successful modes: a merge/cut-in scene, a pedestrian crossing scene, and an unprotected turn scene. For each case, we show the current structured scene state, instantiated with SceneSummary in the present implementation, together with the candidate trajectories, simplified future interaction/occupancy cues derived from rollout, the final risk score for each candidate, and the entity with the largest counterfactual contribution to the selected risk.

These successful cases make the quantitative advantages more interpretable. In the merge/cut-in case, CoordiWorld suppresses the aggressive merge because the predicted interaction with *veh_A* dominates the future risk. In the pedestrian-crossing case, the selected plan yields or stops at the crossing point because the future occupancy of *ped_01* raises both collision risk and predictive uncertainty. In the unprotected-turn case, the final choice is determined jointly by interaction timing and map-grounded constraints rather than by trajectory smoothness alone. Across all three examples, the selected trajectory is the one with the lowest calibrated evaluator score rather than merely the shortest or most aggressive candidate. The entity-level attributions are particularly useful for safety analysis and debugging because they expose why alternative candidates are rejected.

Figure 5 presents three illustrative structured-scene failure modes: a missed occluded cyclist caused by missing upstream evidence, an ambiguous local stop-line/control interpretation, and a late conflict that emerges beyond the effective rollout horizon. Red cross or warning markers denote the selected yet undesirable candidate in each case. Reporting such cases is important because it clarifies the current operating boundary of the method.

TABLE VIII
CLOSED-LOOP TRANSFER STUDY ON BENCH2DRIVE. THIS EXPERIMENT IS INTENDED TO VERIFY DOWNSTREAM RELEVANCE RATHER THAN TO REDEFINE THE PAPER AS A FULL END-TO-END DRIVING SYSTEM. HIGHER IS BETTER. VALUES REPORT THE MEAN OVER THREE RANDOM SEEDS.

Method	DS↑	SR↑
TTC + map-rule	54.2	23.1
Current-state scorer	57.6	25.9
Latent/BEV evaluator-style	59.1	28.0
WoTE-style	60.2	29.4
CoordiWorld (ours)	**61.0**	**30.1**

Although CoordiWorld substantially improves decision-time evaluation, Fig. 5 shows that failure still arises when key actors are missing from the structured scene state, when map-grounded rule context is locally ambiguous, or when the decisive interaction emerges beyond the 4 s rollout horizon. These cases should therefore be interpreted as boundary conditions of the current evaluator design rather than as contradictions of the main quantitative trend.

G. Closed-Loop Transfer to Bench2Drive

In the Bench2Drive transfer study, the proposal generator, the low-level controller, and the simulator interface are kept fixed across all methods; only the evaluator used for candidate selection is replaced. This design isolates the downstream effect of improved trajectory ranking from the rest of the execution stack.

As shown in Table VIII, CoordiWorld improves Driving Score from 60.2 to 61.0 and Success Rate from 29.4 to 30.1 relative to WoTE-style. The margin is naturally smaller than that on pseudo-simulation because the final closed-loop outcome also depends on the proposal generator, the controller, and simulator interaction. This experiment should therefore be viewed as a downstream relevance check rather than as the primary evidence of the paper.

Overall, the closed-loop transfer result supports the claim that a stronger structured evaluator can serve as a better decision module when embedded into an executable control stack.

V. CONCLUSION

This article presented CoordiWorld, a provenance-aware structured world model for auditable fixed candidate-set trajectory evaluation in autonomous driving. Rather than treating world modeling as unrestricted future synthesis, CoordiWorld rolls a conflict-resolved structured scene state forward under candidate ego trajectories and transforms the imagined future into calibrated collision risk, map-grounded rule-violation risk, predictive uncertainty, and post-hoc entity-level attribution. In this way, the framework recasts trajectory evaluation from opaque latent-space scoring into a structured and auditable form of future-aware decision support.

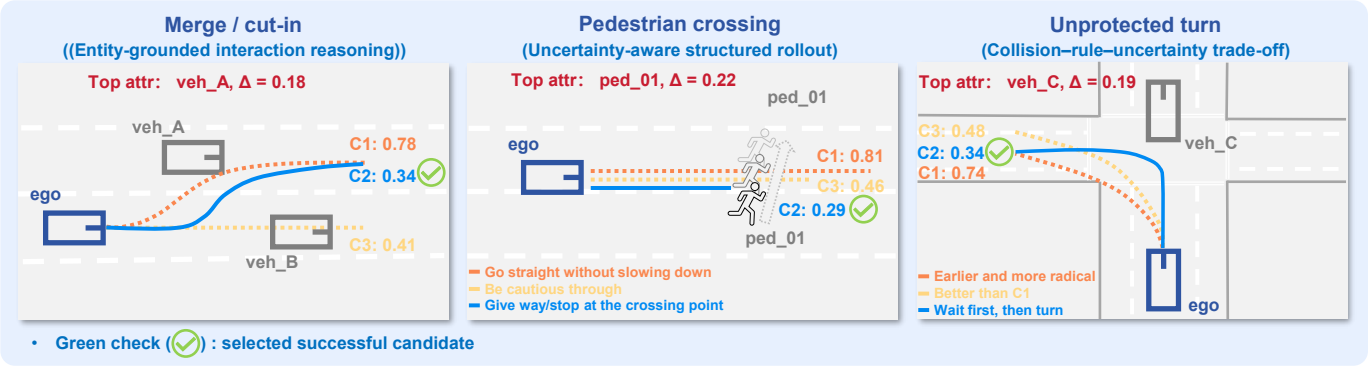

Fig. 4. Qualitative successful cases shown as illustrative structured-scene visual summaries. From left to right: merge/cut-in, pedestrian crossing, and unprotected turn. Each panel shows the candidate trajectories, final per-candidate risk scores (lower is better), the selected candidate (green check), and the top risk-attributed entity under counterfactual masking. In the pedestrian-crossing panel, the repeated pedestrian icons denote future occupancy of the same risky pedestrian across rollout steps rather than multiple distinct pedestrians.

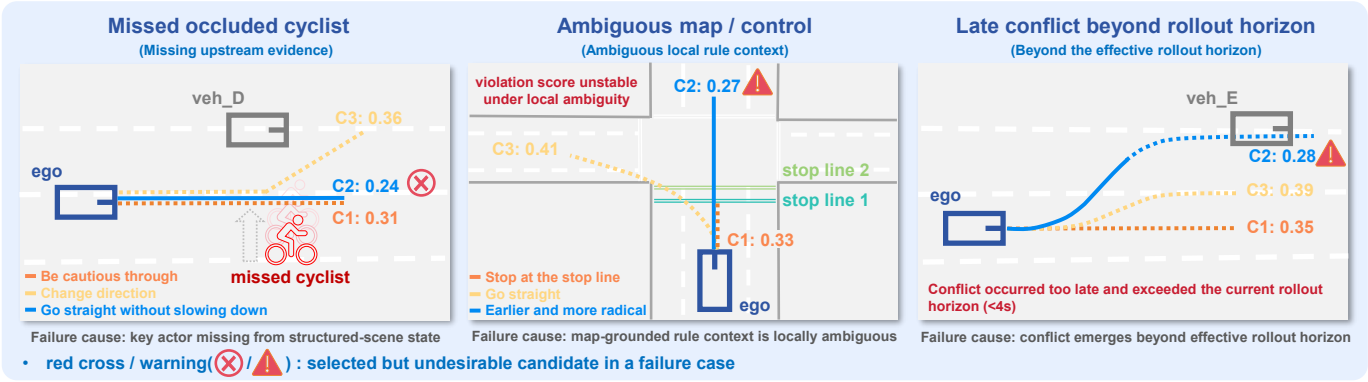

Fig. 5. Qualitative failure modes shown as illustrative structured-scene visual summaries. From left to right: missing upstream evidence (missed occluded cyclist), ambiguous local map/control context, and conflict emerging beyond the effective rollout horizon. Red cross or warning markers denote the selected but ultimately undesirable candidate. These examples clarify that the current formulation remains sensitive to upstream omissions, local rule-context ambiguity, and finite-horizon rollout.

Experimental results on the NAVSIM-v2-aligned shared-candidate evaluator benchmark showed that CoordiWorld achieves the strongest overall EPDMS among same-protocol baselines and improves the strongest comparator by 1.1 points. The gains are concentrated on aspects related to safety and stability rather than ego progress alone, and are accompanied by better rank correlation, lower top-1 selected-plan collision and violation rates, lower calibration error, stronger robustness under structured-input perturbations, and stronger auditability metrics such as Risk Drop@1 and Entity Recall@3. A small-scale closed-loop transfer study on Bench2Drive further suggests that better candidate evaluation can translate into better executable control behavior.

Taken together, these findings indicate that structured and entity-grounded future evaluation can serve as a practical building block for trustworthy decision support in connected and cyber-physical driving systems. The current formulation, however, still depends on the quality of upstream scene evidence, remains sensitive to local ambiguity in map geometry and traffic-control context, and is bounded by a finite rollout horizon. Future work will therefore focus on tighter integration with upstream evidence management and downstream control, more adaptive reasoning under partial observability, and longer-horizon structured evaluation for safety-critical autonomous driving.

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